Skill, risk and luck in feedback-guessing games

The Lexodle project

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Abstract

We define skill, risk and luck for a guess in a word game with colour feedback (Wordle and variants). The answer is drawn by a known *law*: every word equally likely, or each by its weight (word frequency). A guess partitions the words still possible by the feedback they would give. The answer lies in one class; the *pool* is the class's share of the law, expressed as a number of equally likely words (its size, under equal chances). *Skill* is the log-reduction of the expected pool, as a fraction of the same quantity for the best real guess. *Luck* is the probability of a larger pool minus the probability of a smaller one. It lies strictly in $(-100, 100)$ percent. Under the law the answer is drawn from, and only under that law, it has mean zero for every guess, and its increments over a game are martingale differences, so luck is uncorrelated with the skill of the same and earlier guesses (not independent of it). The two simpler luck definitions are unbounded or biased. The log-reduction of the candidate set equals skill plus risk plus luck exactly, where risk is the gap between Shannon and collision entropy. On the last try skill is the guess's chance of winning as a share of the best chance any guess has, and luck is the outcome less that chance. *Daring* measures how far the player went for the whole word instead of the clues: it is zero for a guess that cannot be the answer, and otherwise grows with what the attempt cost in skill, how unlikely the word was, and how many tries were left. At the opening guess, where every word is possible, it is a function of skill alone (and the word's chance). We extend the definitions to games with several boards, give an exact algorithm, and report experiments on an English five-letter list ($N = 4{,}994$) and simulated players of several habits. The best first guess leaves 114.4 words on average (equal chances). CRANE scores 81.6 (97.6 on a linear scale, which puts most openings near 100). A bound on the best conceivable split from the number of feedback patterns is too loose to use. We list open problems.

1 Introduction

In Wordle-like games the player guesses a word. Each letter is marked as in the right place, elsewhere in the word, or absent. After the game two questions arise. Skill: given what was known, how good was the guess? Luck: given the guess, how favourable was the feedback? A score that mixes the two is uninformative.

Entropy-based solvers choose the guess that maximises the expected information in the feedback [1, 6], as for the code game Mastermind [2, 3]. Exact methods give optimal strategies for the original word list [4]. For generalised word lists, deciding whether a win within a given number of guesses can be guaranteed is NP-hard [5]. A newspaper review tool reports skill and luck per guess [7]; we know of no published formula. This paper does not address how to play. It defines scores to show a player after the game. The requirements are:

(R1) **Knowledge only.** Neither score depends on the answer except through what the player saw (and the law the game draws answers by, which is a rule of the game).

(R2) **Bounded.** Skill runs from 0 (the guess tells nothing) to 100 (as good as any real guess). Luck is symmetric about 0 and bounded.

(R3) **Fair luck.** Under the law the answer is drawn from, luck has mean zero (given the history) and is uncorrelated with the skill of the same and earlier guesses. It is unbiased only with respect to that law (Section 2.1).

(R4) **One plot.** Both scores can be read from a single plot of the outcomes the guess could have had.

(R5) **Accounting.** The reduction of the candidate set equals a skill term plus a risk term plus a luck term, exactly (Section 5).

Section 2 fixes the setting, Section 3 defines skill, Section 4 luck, Section 5 the accounting, Section 6 daring, Section 7 several boards, Section 8 the algorithm, Section 9 the experiments, Section 10 the open problems. Code and scripts are listed in Section 11.

2 Setting

Notation. ℓ is the word length, $N = |\Omega|$ the number of words still possible, W the (random) answer, X the pool a guess leaves (a number of words, possibly effective, see below), $S \in [0, 1]$ a skill and $D \in [0, 1)$ a daring (both quoted as percentages), $\Lambda \in (-1, 1)$ a luck (quoted as a percentage).

Let Ω be the finite set of words still possible given everything the player has seen, $N = |\Omega| \geq 1$, and $\mathcal{G}$ the finite set of admissible guesses. Assume $\Omega \subseteq \mathcal{G}$. Let Φ be the set of feedback patterns ($\{\text{grey}, \text{yellow}, \text{green}\}^\ell$ for ℓ-letter words) and $f\colon \mathcal{G} \times \Omega \to \Phi$ the feedback function. Only two properties of f are used: it is computable, and $f(g, w)$ is all green if and only if $g = w$.

2.1 Answer laws

Definition 1 (Answer law)**.** A law is a weight $\pi(\omega) > 0$ for each $\omega \in \Omega$, $M = \sum_{\omega \in \Omega} \pi(\omega)$. The answer W has $\mathbb{P}(W = \omega) = \pi(\omega)/M$. Feedback is a deterministic function of the answer and the guesses, so given the history of guesses and feedback, W has the law π restricted to the words consistent with the history, and M is replaced by the weight of those words.

Three laws are used. **(U)** Every possible word equally likely, $\pi \equiv 1$ (frequency off). **(F)** Each word by its weight, its frequency in the word list (frequency on). **(D)** The daily: uniform over the 3,000 most common words and zero elsewhere. A word with zero chance is not possible, so Ω is those 3,000 words and (D) is (U) on a smaller Ω; the other words can be played as probes ($\mathcal{G} \supsetneq \Omega$). This is how `predict_tries3.dart` treats it. Every score below is relative to one law, which is part of the rules of the game and not of the answer, so (R1) holds. What each score is relative to is stated in Theorem 3 and Section 4.1.

A guess $g \in \mathcal{G}$ partitions Ω into nonempty classes $C_\varphi = \{w \in \Omega : f(g, w) = \varphi\}$, k of them. Let $\mu_i = \pi(C_i)$ be the mass of class i and $q_i = \mu_i/M$ the chance that the answer lies in it,

$\sum_i q_i = 1$. The *effective size* of the class is $s_i = Nq_i$, so $\sum_i s_i = N$; under (U) $s_i = |C_i|$. The answer lands in one class. Its effective size $X_g = s_{i(W)}$ is the *pool*: the number of equally likely words that would be as hard to tell apart. Under (U) it is the number of words still possible after the guess. If ν_x classes have effective size x,

$$\mathbb{P}(X_g = x) = \frac{x\,\nu_x}{N}. \tag{1}$$

Pools are size-biased: a large class is more likely to contain the answer. The definitions of skill and luck, Lemma 1, Proposition 2 and Theorems 3 and 4 use only N, the effective sizes and their order, and hold for every law with sizes read as effective sizes (which need not be integers). Statements that count words, such as the examples of Proposition 5, are for (U).

3 Skill

Definition 2 (Expected pool)**.** $E(g) = \mathbb{E}[X_g] = \dfrac{1}{N}\sum_{i=1}^{k} s_i^2 = N\sum_{i=1}^{k} q_i^2$.

Under (U) this is $\sum_i |C_i|^2/N$. In general $\sum_i q_i^2$ is the chance that two independent answers fall in the same class, and E is N times it.

Definition 3 (Best real guess, skill)**.** $E^* = \min_{g\in\mathcal{G}} E(g)$. For $N \geq 2$ the *skill* of $g \in \mathcal{G}$ is

$$S(g) = \frac{\ln\bigl(N/E(g)\bigr)}{\ln\bigl(N/E^*\bigr)} \in [0,1],$$

reported as a percentage. For $N = 1$, $S(g) = 1$ if g is the remaining word and 0 otherwise.

The logarithm is used because a guess cuts the candidate set by a factor. On the linear scale $(N - E)/(N - E^*)$ most sensible guesses score near 100 (Section 9). The linear score has the same endpoints and the same ordering of guesses.

Lemma 1. *Let the partition of g have k classes, and let $E_{\min} = N\sum_{\omega\in\Omega}(\pi(\omega)/M)^2$ (the value of E for the partition into singletons).*

1. *$\max\{N/k, E_{\min}\} \leq E(g) \leq N$; $E(g) = N$ iff $k = 1$; $E(g) = E_{\min}$ iff $k = N$; $E(g) = N/k$ iff all classes have equal mass. Also $E_{\min} \geq 1$, with equality iff π is constant. Under (U), $E(g) = 1$ iff $k = N$.*

2. *If the partition of g' properly refines that of g, then $E(g') < E(g)$.*

Proof. (1) $q_i \leq 1$ gives $\sum q_i^2 \leq \sum q_i = 1$, with equality iff $k = 1$, so $E \leq N$. By Cauchy–Schwarz $\sum q_i^2 \geq (\sum q_i)^2/k = 1/k$, with equality iff the q_i are equal. A class of chance q_i is a sum of the chances $p_\omega = \pi(\omega)/M$ of its words, and $(\sum p_\omega)^2 \geq \sum p_\omega^2$, with equality iff the class has one word (the weights are positive); so $E \geq E_{\min}$, with equality iff $k = N$. Finally $\sum p_\omega^2 \geq 1/N$ by Cauchy–Schwarz, with equality iff the p_ω are equal. (2) Splitting a class of chance $b_1 + \cdots + b_r$, $r \geq 2$, replaces $(\sum b_j)^2$ by $\sum b_j^2$, which is smaller because the cross terms are positive. □

Proposition 2. *Let $N \geq 2$ and $\Omega \subseteq \mathcal{G}$. Then $E^* < N$, so S is defined, and*

1. $0 \le S(g) \le 1$ *for every* $g \in \mathcal{G}$*;* $S(g) = 1$ *iff* $E(g) = E^*$*;* $S(g) = 0$ *iff* g *does not divide* Ω*;*

2. S *is strictly decreasing in* $E(g)$ *and affine in* $\ln E(g)$*: halving the expected pool adds* $\ln 2 / \ln(N/E^*)$*. If the partition of* g' *properly refines that of* g*, then* $S(g') > S(g)$*;*

3. $S(g)$ *is a function of* $(\Omega, \pi, g, \mathcal{G})$ *alone. It does not depend on the answer.*

Proof. Take $w \in \Omega \subseteq \mathcal{G}$. The guess w gives all green against w, and against no other word (all green iff $g = w$), so $\{w\}$ is a class of w's partition. For $N \ge 2$ the partition is nontrivial, so $E(w) < N$ by Lemma 1 and $E^* < N$. The denominator is positive. (1) follows from $E^* \le E(g) \le N$ and the lemma, (2) from the formula and Lemma 1(2), (3) because $E(g)$ and E^* are computed from the classes of Ω. □

Remark 1 (Entropy). Then $\ln(N/E) = -\ln \sum q_i^2 = H_2$, the Rényi entropy of order 2 [9] of the split chances; e^{H_2} is the inverse Simpson index, the Hill number of order 2 [8, 10]. Hence $S(g) = H_2(g) / \max_{g'} H_2(g')$: the collision-entropy share of the best guess. The Shannon entropy $H = -\sum q_i \ln q_i = \ln N - \mathbb{E}[\ln X_g]$ [1] is the expected log-reduction of the candidate set, and $H \ge H_2$ by Jensen. H_2 is used because it is a function of E, the expected number of words left. Other orders of Rényi entropy give other scales (Section 9.4).

Remark 2 (Why the best real guess). A bound that needs no search is $e = N / \min(N, 3^\ell)$, the most even split into $\min(N, 3^\ell)$ groups (a lower bound on E under every law, by Lemma 1(1)). Section 9 tests it and two bounds that use the clues; they never improve on it, and it is too loose. The best real word makes about two thirds of the groups the bound allows when many words remain, and rarely separates every word when few remain. E^* is therefore the actual minimum over real guesses. The floor of E^* is $E_{\min}$ (Lemma 1), reached only if some guess separates every word.

4 Luck

Let $q_x = \mathbb{P}(X_g = x) = x\,\nu_x / N$. Pools are ordered by effective size, that is by the chance of their class: a class is bigger when it is more likely to hold the answer.

Definition 4 (Luck)**.** For x in the support of $X = X_g$,

$$\Lambda_g(x) = \mathbb{P}(X > x) - \mathbb{P}(X < x).$$

The luck of an outcome is $\Lambda_g(X_g)$, reported as a percentage.

It is the probability of a larger pool minus the probability of a smaller one: negative if the outcome is worse than most, positive if better, zero at the median.

Theorem 3. *Let the answer be drawn by the law* π *used to compute the effective sizes, let* X *have the law* (1) *and* $\Lambda = \Lambda_g$*. (Only the law of* X *is used in the proof; (U) is the case* $s_i = |C_i|$*.)*

1. *(Mid-distribution form.)* $\Lambda(x) = 2\tilde{F}(x) - 1$ *with* $\tilde{F}(x) = \mathbb{P}(X > x) + \frac{1}{2}\,\mathbb{P}(X = x)$*.* $\Lambda(X)$ *is Cliff's delta between the realised pool and an independent copy [14].*

2. *(Bounded.)* $-1 < \Lambda(x) < 1$ *for every* x *in the support, and* Λ *is strictly decreasing in* x*.*

3. *(Fair.)* $\mathbb{E}[\Lambda(X)] = 0$ *for every guess, when the answer follows the law* π*.*

4. *(Spread.)* $\operatorname{Var}\Lambda(X) = \dfrac{1-\sum_x q_x^3}{3} < \dfrac{1}{3}$*, the sum over the distinct pool sizes* x *(equal pools are one atom:* q_x *is the chance of landing in any class of effective size* x*, so ties lower the variance).*

5. *(Invariance.)* $\Lambda(x)$ *is unchanged if all class sizes (under (U)) or all weights (in general) are multiplied by the same positive factor. Equal pools have equal luck.*

Proof. (1) $\mathbb{P}(X < x) = 1 - \mathbb{P}(X > x) - \mathbb{P}(X = x)$, so $\Lambda(x) = 2\,\mathbb{P}(X > x) + \mathbb{P}(X = x) - 1 = 2\tilde{F}(x) - 1$. For an independent copy X', $\Lambda(x) = \mathbb{E}[\mathbf{1}\{X' > x\} - \mathbf{1}\{X' < x\}]$, which is Cliff's delta.

(2) $\mathbb{P}(X > x) + \mathbb{P}(X < x) = 1 - q_x < 1$ and both terms are non-negative, so $|\Lambda(x)| < 1$. If $x < y$ are in the support, $\mathbb{P}(X > x) \geq \mathbb{P}(X > y) + q_y$ and $\mathbb{P}(X < x) \leq \mathbb{P}(X < y) - q_x$, so $\Lambda(x) > \Lambda(y)$.

(3) For independent identically distributed X, X', $\mathbb{E}[\Lambda(X)] = \mathbb{P}(X' > X) - \mathbb{P}(X' < X) = 0$ by exchangeability.

(4) Order the distinct pool sizes $x_1 > \cdots > x_r$ with masses q_i and $c_i = \sum_{j<i} q_j = \mathbb{P}(X > x_i)$, so $\tilde{F}(x_i) = c_i + q_i/2$. The sum $\sum_i[(c_i + q_i)^3 - c_i^3] = 1$ telescopes, so

$$\sum_i \left(3c_i^2 q_i + 3c_i q_i^2 + q_i^3\right) = 1, \quad \text{hence} \quad \sum_i q_i c_i^2 + \sum_i c_i q_i^2 = \frac{1-\sum_i q_i^3}{3}.$$

Then $\mathbb{E}[\tilde{F}^2] = \sum_i q_i(c_i + q_i/2)^2 = \sum_i q_i c_i^2 + \sum_i c_i q_i^2 + \frac{1}{4}\sum_i q_i^3 = \frac{1}{3} - \frac{1}{12}\sum_i q_i^3$. By (3), $\mathbb{E}[\tilde{F}] = \frac{1}{2}$, so $\operatorname{Var}\tilde{F} = \frac{1}{12}(1 - \sum_i q_i^3)$ and $\operatorname{Var}\Lambda = 4\operatorname{Var}\tilde{F}$.

(5) Multiplying every weight (every size, under (U)) by $c > 0$ multiplies each mass μ_i and M by c, so each chance q_i and hence each q_x is unchanged. □

Remark 3. Λ is the mid-distribution transform of the pool [11, 12], a discrete analogue of the probability integral transform [13]. Giving tied pools their mid-rank makes the mean exactly zero. If columns of a plot are as wide as their chance, the horizontal middle is the median outcome and Λ is the position of the landed column relative to that middle (Figure 1).

4.1 Luck under the wrong law

Theorem 3(3) is a statement about one law: the one that makes the class chances q_i and the one the answer is drawn from. If the answer follows another law π', with class chances q'_i, the mean of Λ_g is $\sum_i q'_i \Lambda_g(s_i)$, which is not zero in general (Section 9.8: luck computed with equal chances has a mean of +10.6% for CRANE when answers are drawn by frequency). So luck is unbiased with respect to π only; under (U) it is unbiased only for equal chances, and under (F) only for frequency. The app computes luck, skill and daring with the law the game draws by, so all three use one law.

Ties get their mid-rank. Suppose a tie rule gives tied pools weight $\lambda \in [0, 1]$: $\Lambda^\lambda(x) = \mathbb{P}(X > x) - \mathbb{P}(X < x) + (2\lambda - 1)\,\mathbb{P}(X = x)$, so $\lambda = \frac{1}{2}$ is the mid-rank, $\lambda = 0$ counts ties as smaller (strict ranks) and $\lambda = 1$ as larger. With q_i the chances of the distinct pool sizes,

$$\mathbb{E}[\Lambda^\lambda(X)] = (2\lambda - 1)\sum_i q_i^2,$$

because $\mathbb{E}[\mathbb{P}(X' = X)] = \sum_i q_i^2$ for an independent copy X' and the $\lambda = \frac{1}{2}$ term is zero by Theorem 3(3). Since $\sum_i q_i^2 > 0$, in this family of tie rules only the mid-rank has mean zero, and it does for every law and every guess.

4.2 Luck is uncorrelated with skill

Let $\mathcal{F}_{t-1}$ be the history (guesses and feedback) before guess t. The guess g_t is a function of $\mathcal{F}_{t-1}$ and of a randomisation ξ_t independent of the answer and of everything before it. Let $\mathcal{G}_t = \sigma(\mathcal{F}_{t-1}, \xi_t)$, so that g_t is $\mathcal{G}_t$-measurable and $\mathcal{F}_{t-1} \subseteq \mathcal{G}_t$. Let S_t and $\Lambda_t = \Lambda_{g_t}(X_{g_t})$ be its skill and luck, with the convention that $\Lambda_t = 0$ and $S_t = 0$ once the game has ended. After the answer was found there is one word left and one class, so Theorem 3 gives luck 0 in any case; after the last permitted try there is no guess and the convention applies.

Theorem 4. *Let the answer follow the law π used to score. Then $\mathbb{E}[\Lambda_t \mid \mathcal{G}_t] = 0$, hence $\mathbb{E}[\Lambda_t \mid \mathcal{F}_{t-1}] = 0$ and $\mathbb{E}\,\Lambda_t = 0$. Moreover $\operatorname{Cov}(\Lambda_t, S_s) = 0$ for $s \le t$, $\operatorname{Cov}(\Lambda_s, \Lambda_t) = 0$ for $s \ne t$, and $\sum_{s \le t} \Lambda_s$ is a martingale for the filtration $(\mathcal{F}_t)$. The last-try luck of Proposition 6 has the same conditional mean zero.*

Proof. Given $(\mathcal{F}_{t-1}, \xi_t)$, ξ_t being independent of W, the answer has the law π restricted to the consistent words Ω_t, and g_t is fixed; the classes and chances of g_t are those used in the score. Theorem 3(3) applies, so $\mathbb{E}[\Lambda_t \mid \mathcal{G}_t] = 0$. The tower property gives $\mathbb{E}[\Lambda_t \mid \mathcal{F}_{t-1}] = 0$. For $s \le t$, S_s is $\mathcal{G}_t$-measurable (it is a function of Ω_s and g_s, and for $s < t$ both are in $\mathcal{F}_{t-1}$), and Λ_s for $s < t$ is a function of g_s and the feedback, hence $\mathcal{F}_{t-1}$-measurable. All are bounded. So $\operatorname{Cov}(\Lambda_t, S_s) = \mathbb{E}[S_s\,\mathbb{E}[\Lambda_t \mid \mathcal{G}_t]] - \mathbb{E}\,S_s\,\mathbb{E}\,\Lambda_t = 0$, and likewise for $\Lambda_s\Lambda_t$. The partial sums are adapted to $(\mathcal{F}_t)$ and have increments of conditional mean zero. On the last try the conditional mean of the win indicator given $\mathcal{G}_t$ is the win chance (Proposition 6), so the same argument applies. □

This is (R3). It does *not* say that luck is independent of skill (the spread of Λ_t depends on the split chosen), and it does not say that the game's total luck is uncorrelated with its total skill: $\operatorname{Cov}(\Lambda_t, S_s)$ for $s > t$ need not vanish, since a lucky early outcome changes the position and so what later skills measure. The regression of Section 9.6 uses game means, to which only the case $s \le t$ applies.

4.3 Alternative luck definitions

Two definitions scale the difference between the landed pool a and the expected pool E. Pools are class sizes under (U) here, which is the law of the comparison in Section 9.

Proposition 5. *Let $E > 1$.*

1. *The expectation-scaled luck $\Lambda^{\exp}(a) = (E-a)/(E-1)$ has mean zero and no lower bound: for one pair and $N-2$ singletons ($N \ge 3$), landing in the pair gives $-(N-2)/2$.*
2. *The rescaled luck $\Lambda^{\mathrm{res}}(a) = (E-a)/(E-1)$ for $a \le E$ and $(E-a)/(a_{\max}-E)$ for $a > E$ lies in $[-1,1]$. If the pools are not all equal, its mean is $\kappa\big(\frac{1}{E-1} - \frac{1}{a_{\max}-E}\big)$, $\kappa = \mathbb{E}[(E-X)^+] > 0$, which is nonzero unless $a_{\max} - E = E-1$. For the partition $\{3,1\}$ it is $-\frac{1}{2}$. If all classes have equal size, $X \equiv E$, $\Lambda^{\mathrm{res}} \equiv 0$ and the mean is 0 (for $\{2,2\}$, $E = 2$).*

Proof. (1) $\mathbb{E}\,\Lambda^{\exp} = (E - \mathbb{E}\,X)/(E-1) = 0$. For a pair and $N-2$ singletons $E = (N+2)/N$, so $E - 1 = 2/N$, $E - 2 = -(N-2)/N$, and the ratio is $-(N-2)/2$. (2) If X is not constant then $1 < E < a_{\max}$ and $\kappa > 0$. $\mathbb{E}[X - E] = 0$ gives $\mathbb{E}[(E-X)^+] = \mathbb{E}[(X-E)^+] = \kappa$. The positive part of Λ^{res} is scaled by $1/(E-1)$ and the negative part by $1/(a_{\max} - E)$. For $\{3,1\}$: $N = 4$, $E = \frac{5}{2}$, $a_{\max} = 3$. Landing in the 3 (chance $\frac{3}{4}$) gives -1, landing in the 1 gives $+1$, mean $-\frac{1}{2}$. If $X \equiv E$ every landed pool has $a = E$ and $\Lambda^{\mathrm{res}} = 0$. □

The first violates (R2), the second (R3): its bias mixes guess quality into luck. Section 9 gives means from +19% to −60% on real positions.

4.4 The last try

On the last guess only a word that fits the clues can win, and the split is worth nothing, so the split skill of Section 3 no longer measures the guess. Let N words fit, with law π and total weight M. A guess $g \in \Omega$ wins with chance $p(g) = \pi(g)/M$ (equal to $1/N$ under (U)); any other guess cannot win, $p(g) = 0$. Let $p^* = \max_{h \in \Omega} p(h)$.

Definition 5 (Last-try skill and luck)**.** For $N \geq 2$ the *skill* of the last guess is $S^{\mathrm{last}}(g) = p(g)/p^*$: the chance of winning as a share of the best chance any guess has. Its *luck* is $\Lambda^{\mathrm{last}} = Y - p(g)$, with $Y = \mathbf{1}\{\mathrm{win}\}$.

Proposition 6. *Under the law the answer is drawn from,* $\mathbb{E}[\Lambda^{\mathrm{last}}] = 0$ *for every guess (given the history and the guess).* Λ^{last} *takes the values* $1 - p$ *and* $-p$*; for* $N \geq 2$ *it lies in* $(-1, 1)$*, and it is* 0 *for a guess that cannot win.* $S^{\mathrm{last}} \in [0,1]$*, is* 1 *iff* g *is a most likely word and* 0 *iff* g *is a probe, and depends only on the position and the guess. Under (U),* $S^{\mathrm{last}} = 1$ *for every word that fits. For every guess*

$$Y = S^{\mathrm{last}}(g)\, p^* + \Lambda^{\mathrm{last}},$$

the part of the outcome the guess had in hand plus the part that fell by chance. A probe has skill 0*: on the last try the loss from playing a word that cannot win is in skill. With three words fitting under (U), a hit is* +67% *and a miss* −33%*.*

Proof. $\mathbb{E}\,Y = p$, so $\mathbb{E}[Y - p] = 0$. If $N \geq 2$ and $p > 0$ then $0 < p < 1$ (the weights are positive and there is another word), so $1 - p \in (0,1)$ and $-p \in (-1, 0)$; if $p = 0$ then $Y = 0$ and the luck is 0. $0 \leq p \leq p^*$ gives the range of S^{last}, with equality as stated because $p(g) = 0$ iff $g \notin \Omega$ and $p(g) = p^*$ iff g attains the maximum. The identity is $p = (p/p^*)p^*$ added to $Y - p$. Under (U), $p = 1/N = p^*$ for every $g \in \Omega$. □

The accounting of Section 5 is about the split and does not apply on the last try; the identity above takes its place. The conditional mean zero is what Theorem 4 uses.

Example 1. Twenty words are split into groups of $10, 5, 3, 1, 1$. Then $E = (100 + 25 + 9 + 1 + 1)/20 = 6.8$. If no real guess beats singletons ($E^* = 1$), the skill is $\ln(20/6.8)/\ln 20 = 36.0\%$ ($(20 - 6.8)/19 = 69.5\%$ on the linear scale). Luck by landing: in the 10, $0 - \frac{10}{20} = -50\%$; in the 5, $\frac{10}{20} - \frac{5}{20} = +25\%$; in the 3, $\frac{15}{20} - \frac{2}{20} = +65\%$; in a 1, $\frac{18}{20} - 0 = +90\%$ (the two singletons tie and neither counts against the other). The chance-weighted mean is $\frac{10}{20}(-.5) + \frac{5}{20}(.25) + \frac{3}{20}(.65) + \frac{2}{20}(.9) = 0$.

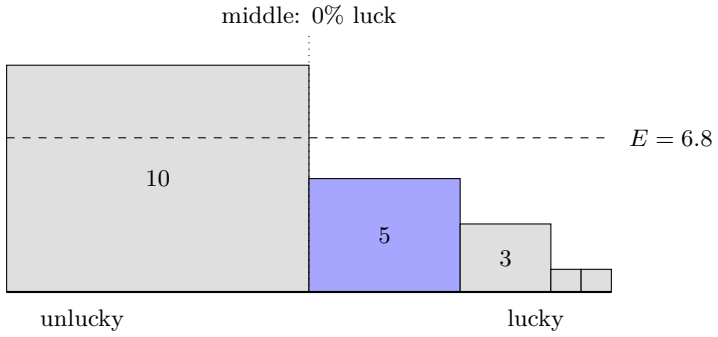


Figure 1: Example 1: twenty words in groups of $10, 5, 3, 1, 1$. Column width is chance, height is pool size. The dashed line is $E = 6.8$. The shaded group of 5 has luck +25%: chance of a larger pool (10/20) minus chance of a smaller one (5/20).

5 Accounting: skill, risk and luck

Skill and luck have different scales, so they do not add. Their underlying quantities do. Let $X = X_g$ be the pool landed in and $\rho = \ln(N/X)$ the *total reduction* of the candidate set (nats; divide by $\ln 2$ for bits): under (U), X words remain of N; in general $\rho = -\ln q$ for the chance q of the class landed in. With $H = -\sum_i q_i \ln q_i$,

$$\underbrace{H_2}_{\text{skill}} + \underbrace{H - H_2}_{\text{risk}} + \underbrace{\rho - H}_{\text{luck}} = \rho, \qquad H_2 = \ln \frac{N}{E}.$$

Proposition 7 (Accounting)**.** *For every guess and outcome (not on the last try, Proposition 6) the three terms sum to ρ, under any answer law. Moreover*

1. *the skill term $H_2 = \ln(N/E) \geq 0$ is the numerator of S;*
2. *the risk term $H - H_2 = \ln \mathbb{E}[X] - \mathbb{E}[\ln X] \geq 0$, with equality iff all classes have equal size;*
3. *the luck term $Z = \rho - H = \mathbb{E}[\ln X] - \ln X$ has $\mathbb{E}[Z] = 0$ (under the law the answer follows) and $\operatorname{Var} Z = \operatorname{Var}(\ln X)$, and under the hypotheses of Theorem 4 it is a martingale difference.*

Proof. With $X = Nq_{i(W)}$, $\mathbb{E}[\rho] = \mathbb{E}[\ln(N/X)] = \ln N - \mathbb{E} \ln X = -\sum_i q_i \ln q_i = H$ and $H_2 = \ln N - \ln E$. Hence $H - H_2 = \ln \mathbb{E}[X] - \mathbb{E}[\ln X] \geq 0$ by Jensen's inequality, with equality iff X is constant, which by (1) holds iff all class sizes are equal. The luck term is $\rho - \mathbb{E}\,\rho$: mean zero, variance $\operatorname{Var} \rho = \operatorname{Var}(\ln X)$. The conditional statement follows as in Theorem 4, applied to $\ln X$. □

Remark 4 (Risk). The risk term is the convexity gap between the average of the log and the log of the average. An even split has $H = H_2$ and zero risk. One large class and several small ones has a large gap: low expected reduction, high payoff if the answer is in a small class. Take 100 words in ten classes of ten, against one class of 91 and nine singletons. The first has $H_2 = H = 3.32$ bits and risk 0. The second has skill 0.27, risk 0.45 and expected reduction 0.72 bits. Landing in a singleton (chance 9%) reduces by 6.64 bits $= 0.27 + 0.45 + 5.92$. Landing in the 91 (chance 91%) reduces by $0.14 = 0.27 + 0.45 - 0.59$.

guess	skill H_2	risk $H - H_2$	expected H	sd of ρ
tares	5.45	0.77	6.22	1.71
slate	5.18	0.78	5.96	1.78
crane	4.44	0.97	5.42	2.02
abbey	3.18	0.70	3.88	1.91
queue	2.04	0.77	2.81	1.99
mamma	1.57	0.84	2.41	2.18

Table 1: The terms for first guesses (bits), $N = 4{,}994$. For CRANE, landing in a 641-word class reduces by 2.96 bits ($= 4.44 + 0.97 - 2.46$), in a 23-word class by 7.76 ($= 4.44 + 0.97 + 2.34$), in a single word by 12.29 ($= 4.44 + 0.97 + 6.87$).

Remark 5 (Two luck scores). $Z = \mathbb{E}[\ln X] - \ln X$ and the percentile luck Λ are both strictly decreasing in the landed pool X and both have mean zero, so they rank outcomes identically. Z has the units of skill and risk, which makes the accounting exact, and is bounded by $\pm \ln N$. Λ is the mid-rank of the pool and is bounded by ± 1.

Table 1 gives the terms for some first guesses.

6 Daring

Risk is a property of a guess. It does not say what the player was trying to do. A guess that could be the answer is an attempt at the word; one that cannot is a probe, chosen for what it tells. *Daring* measures how much a guess was an attempt at the whole word rather than a way of narrowing the pool. Fix a position with $\tau \geq 1$ tries left, this one included, a set Ω of $N \geq 2$ words still possible and the answer law π with total weight M (Section 2.1), the same law that scores skill and luck. Let $S(g) \in [0, 1]$ be the skill of g (the split skill of Section 3; daring is 0 when $\tau = 1$, so the last-try skill is not needed), and

$$p(g) = \frac{\pi(g)}{M} \text{ if } g \in \Omega, \qquad p(g) = 0 \text{ otherwise,}$$

the chance that g wins now. The reference for "going for the word" is a word that is still possible, as drawn by the answer law: the indicator and p compare g with the words the law could have drawn.

Definition 6 (Daring)**.** The daring of a guess g is

$$D(g) = \mathbf{1}[g \in \Omega] \left(1 - S(g)\right) \left(1 - p(g)\right) \left(1 - \frac{1}{\tau}\right) \in [0, 1),$$

quoted in percent. It is undefined when only one word is possible.

The four factors say, in turn: the guess is a try at the word; it gave up some of the splitting the best guess would have done; the word was unlikely; and there were tries to spare. With one try left going for the word is the only play, so $D = 0$. A guess that is both the best split and a possible answer costs nothing, so $D = 0$ too.

Proposition 8. *1. $0 \le D(g) < 1$, and it depends on the position, the law, the choice of g and τ only, not on the answer, so (R1) holds.*

2. $D(g) = 0$ if $g \notin \Omega$, if $\tau = 1$, or if $S(g) = 1$. (The case $p(g) = 1$ would need every other possible word to have chance 0; the weights are positive, so it means $N = 1$, where D is undefined and the case does not arise.)

3. As a function of (S, p, τ), D is non-increasing in S and in p and non-decreasing in τ, the others fixed.

4. Let $g \in \Omega$ and $\tau \ge 2$. Then $p(g) \in (0,1)$ and $S(g) > 0$, so $D(g) < (1-\frac{1}{\tau})(1-p(g)) < 1-\frac{1}{\tau}$: the value $1 - 1/\tau$ is never attained. More precisely $S(g) \ge s_0(p) = \ln\big(1/(1-2p(1-p))\big)/\ln(N/E^*)$ with $p = p(g)$, so $D(g) \le \big(1 - \min\{1, s_0(p)\}\big)(1-p)(1-\frac{1}{\tau})$.

5. At the opening guess every word is possible. If Ω is the whole list and g is in it, then $\mathbf{1}[g \in \Omega] = 1$ and $D(g) = (1 - S(g))(1 - p(g))(1 - \frac{1}{\tau})$, with $\tau = 6$ in the classic game, so $1 - \frac{1}{\tau} = \frac{5}{6}$. Under (U), $p = 1/N$ is the same for every guess and D is a function of skill alone, $D = \frac{5}{6}(1 - \frac{1}{N})(1 - S)$. Under (F), D lies between $\frac{5}{6}(1 - p^*)(1 - S)$ and $\frac{5}{6}(1 - S)$, $p^* = \max_h p(h)$.

6. D does not involve risk, so the accounting of Proposition 7 is unaffected.

Proof. (1)–(3) are read off the product: each factor lies in $[0,1]$, the last is below 1, each is monotone in the stated variable, and the factors do not mention the answer. (4) $g \in \Omega$ and the weights are positive, so $p(g) > 0$, and $p(g) < 1$ since $N \ge 2$. The class of g is $\{g\}$ (all green iff equal), of chance p, and the other classes share chance $1 - p$, so $\sum_i q_i^2 \le p^2 + (1-p)^2 = 1 - 2p(1-p) < 1$. Hence $E(g) < N$, $S(g) > 0$, and $\ln(N/E(g)) = -\ln \sum_i q_i^2 \ge -\ln(1 - 2p(1-p))$; divide by $\ln(N/E^*)$. (5) $\mathbf{1}[g \in \Omega] = 1$ for every g in the list Ω; the rest is substitution, with $0 < p \le p^*$ and $p = 1/N$ under (U). □

At the opening, then, daring tells no more than skill does: CRANE ($S = 0.816$) has $D = 15.3$ and an opener of median skill ($S = 0.674$) has $D = 27.2$, both under (U) and $\tau = 6$. Daring separates players by what they try only through the guesses after the opener.

Remark 6. Daring is reported beside the accounting and is not a term of it. An earlier definition counted, among the guesses of higher skill, the share that were less risky; it mostly measured how far a guess was from the best one, so any player who was not optimal looked daring, and it ignored the tries left. It is replaced because the question a player means by "bold" is whether they went for the word. Skill is relative to the language and the word length, so D is too, through S. The chance p uses the same law as skill and luck. Daring is defined for single-board games: the quartet and the relay race of Section 7 have no daring score.

7 Several boards

Two variants hide several words. In a *quartet* one guess is played on every unsolved board at once. In a *relay race* boards are played in turn, and every earlier guess is applied to a board when it opens.

A guess is made for live boards $b = 1, \dots, B$ with possible sets Ω_b of sizes N_b. Skill S_b and luck Λ_b are computed per board as above, with E_b^* the best real guess for that board and every

board scored by the same law π (the weights are threaded through to each board, restricted to its candidates). Let $w_b = \log_2 \max\{N_b, 2\}$, which depends only on the history, and

$$S = \frac{\sum_b w_b S_b}{\sum_b w_b}, \qquad \Lambda = \frac{\sum_b w_b \Lambda_b}{\sum_b w_b}.$$

Proposition 9. *$S \in [0,1]$ and $-1 < \Lambda < 1$. Suppose that, given the history and the guess, the answer of each live board has, as a marginal, the law π restricted to that board's candidates, which is the law the board is scored by. Then $\mathbb{E}[\Lambda \mid \mathcal{G}_t] = 0$ and Theorem 4 holds for the combined luck. Independence between boards is not required.*

Proof. Convex combinations stay in $[0,1]$ and $(-1,1)$. The weights are $\mathcal{F}_{t-1}$-measurable, so $\mathbb{E}[\Lambda \mid \mathcal{G}_t] = \sum_b w_b \, \mathbb{E}[\Lambda_b \mid \mathcal{G}_t] / \sum_b w_b = 0$ by Theorem 3(3) on each board, which uses only the marginal law of that board's answer. The remainder of Theorem 4's proof uses only this. □

The hypothesis is about how the game draws the boards' answers. It holds if each board's answer is an independent draw from π (under (U) or (F)). It does not hold exactly if the answers are drawn jointly, for instance all distinct or preferring sets that share few letters: then a board's marginal law given the history is not π restricted to its candidates, and the combined luck need not have mean zero. How far the quartet's draw departs from the hypothesis is not measured here, so the zero-mean property is proved for independent draws only. (The earlier statement, with equal chances on every board, is the case (U) with independent draws.)

Remark 7 (Weights). $\log_2 N_b$ is the number of bits still to find on board b. A board with one word left has weight one: playing that word is rewarded and missing it is penalised; its skill is 1 if the guess is that word and 0 otherwise. Uniform weights would let a nearly solved board dominate. The weights are a design choice.

Remark 8 (Head start). In a relay race a guess on the active board narrows boards not yet open. This credit is not scored. It is reported: the *head start* of a board is the number of words possible on it when it opens, with all earlier guesses applied, against the size of the list. An earlier guess can solve a board as it opens. Assigning this credit to the earlier guesses is open.

8 Computation

Feedback for an ℓ-letter guess is packed into an integer in $\{0, \ldots, 3^\ell - 1\}$. For a guess g, class sizes (or class masses, under a weighted law) are counted in an array indexed by pattern, reset by revisiting only the patterns touched. $E(g) = N \sum q_i^2$ costs one pass, $O(\ell N)$, with no allocation. Skill needs $E^* = \min_g E(g)$, a scan of all $|\mathcal{G}|$ guesses: $O(\ell N |\mathcal{G}|)$, about 1.2×10^8 operations for a first guess on $N = |\mathcal{G}| = 4{,}994$ five-letter words, and fewer as N falls. The scan also gives the skill of every guess, so the rank of the player's guess is free. For several boards, boards sharing the same candidate list (all of them at the start) are scored once. A truncated scan gives E^* too large and skill too high; the app scans every guess.

Luck needs the class masses of the player's guess only: sort the distinct masses and accumulate the chances above and below, $O(k \log k)$ for k classes. The last-try skill needs the largest weight among the words that fit.

guess	rank	E	groups	largest	skill	linear
tares	1	114.4	184	350	100.0	100.0
slate	24	138.0	174	355	95.0	99.5
trace	534	219.8	155	625	82.7	97.8
crane	628	229.3	155	641	81.6	97.6
abbey	3786	550.9	78	1069	58.4	91.1
queue	4907	1218.5	40	2063	37.4	77.4
mamma	4973	1681.5	49	2693	28.8	67.9
xxxix	4994	2828.0	21	3598	15.1	44.4

Table 2: First guesses ranked by E among all 4,994 words.

9 Experiments

All figures use the app's English five-letter list ($N = 4{,}994$; every word is a legal guess and a possible answer). Scripts are in `docs/paper/experiments`; outputs in its `results` directory. The Python experiments (first guesses, luck definitions, simulated players, risk, accounting, worked game, bounds, theorem checks) use law (U). The Dart simulations of Sections 9.5 and 9.6 draw answers by frequency or from the daily set, as stated there.

9.1 First guesses

Table 2 gives E, the number of groups, the largest group and the skill of some openings, with the linear skill $(N - E)/(N - E^*)$ for comparison. The best first guess, TARES, leaves 114.4 words on average.

Over all 4,994 words as first guesses, the median linear skill is 94.3 and 81.1% score above 90 (42.7% above 95). On the logarithmic scale the median is 67.4 and 2.7% score above 90 (0.5% above 95). The linear scale discriminates little among sensible openings.

9.2 Luck definitions

Table 3 gives the luck of landing in the largest pool and the mean luck over all answers, for each definition of Section 4.3. The expectation-scaled luck reaches −208% for the largest pool of TARES. The rescaled luck lies in $[-100, 100]$ but its mean over answers ranges from +18.8% to −21.5% on first guesses, and is −60% in a late position (five words left). The chance-weighted luck has mean 0 to the precision shown.

For CRANE, landing in the 641-word pool (chance 12.8%) gives −87%, in a 328-word pool −47%, in a 23-word pool +71%, in a single word +99%. Landing in a pair among $N - 2$ singletons gives $-(N - 2)/N$ chance-weighted and $-(N - 2)/2$ expectation-scaled: −98% and −4,900% at $N = 100$.

9.3 Simulated players

`play_skill.py` plays 60 games each for a strong player (a good opener, then the consistent word leaving the fewest words) and a casual one (a random common opener, then a random consistent word). Answers are uniform on the list. Guesses are scored on the logarithmic scale.

			largest pool			mean over answers	
guess	E	largest	exp.	resc.	chance	resc.	chance
tares	114.4	350	−208%	−100%	−93%	+18.8%	0.0%
crane	229.3	641	−180%	−100%	−87%	+18.1%	0.0%
abbey	550.9	1069	−94%	−100%	−79%	−1.8%	0.0%
queue	1218.5	2063	−69%	−100%	−59%	−12.7%	0.0%
mamma	1681.5	2693	−60%	−100%	−46%	−21.5%	0.0%

Table 3: Luck of landing in the largest pool (expectation-scaled, rescaled, chance-weighted) and mean luck over all answers. The expectation-scaled mean is zero by construction.

valuation	opener	mean guesses	≤ 3	≤ 4	≤ 5	≤ 6
worst case ($\alpha = \infty$)	aloes	4.01 ± 0.03	26.9	75.3	95.9	99.5
collision ($\alpha = 2$)	tares	3.85 ± 0.03	34.3	82.1	97.0	99.4
Shannon ($\alpha = 1$)	tares	3.83 ± 0.03	36.2	82.5	96.9	99.1
groups ($\alpha = 0$)	tares	3.81 ± 0.03	37.8	82.5	97.0	99.2

Table 4: Greedy player under four valuations, 1,000 games each. Shares in percent. Binomial standard errors: 1.2 to 1.5 points at ≤ 3 and ≤ 4, 0.5 to 0.6 at ≤ 5, 0.2 to 0.3 at ≤ 6 (the same answers are used in every row, so differences between rows are paired and tighter; not computed).

Per guess the strong player's median is 100 (mean 90) and the casual player's 74 (mean 74). In won games the strong player has every guess above 80 in 64% and above 90 in 27%; the casual player in 0% for both. On the linear scale nearly every guess of either player would score above 90.

9.4 Is risk worth rewarding?

H_2 rewards even splits. H (the expected reduction; the risk term is $H - H_2$) credits a split into one large and several small classes more, and counting groups more still. These are the Rényi entropies of order $\alpha = \infty$ (the largest class), 2 (the app), 1 (Shannon) and 0 (number of classes). `risk_attitude.py` plays a greedy player ranking guesses by each. The first guess is chosen over every word; later guesses among the words still possible and a sample of common words. The same 1,000 uniformly drawn answers are used for each (Table 4).

The worst-case valuation is slower (4.01 against about 3.8, four to five standard errors of the difference, treating the runs as unpaired). The other three have indistinguishable mean time. The share won within three guesses rises with the weight on small classes (34.3%, 36.2%, 37.8%), with no loss detectable within six (99.4, 99.1, 99.2%: differences of 0.1 to 0.3 points against a standard error of about 0.3). This is consistent with a gamble paying off when few guesses are needed, but the evidence is weak: one greedy policy, a restricted candidate set, one language and length, one run. The three non-worst-case players share the opener TARES, so their differences come from later guesses.

player	won (%)	guesses	skill	daring
best	100	3.6	100.0	0.0
near-best	100	3.7	99.0	0.3
best-fit	100	3.4	91.4	4.0
frequent	100	3.1	66.3	19.7
random-fit	97	4.3	70.7	21.4
wild	1	6.0	48.2	6.0
wild-bold	0	6.0	46.3	6.5

Table 5: Simulated players, English, five letters, 200 games each. Won within six guesses, guesses per game, and the means of skill and daring.

9.5 Daring and playstyle: simulated players

The app reads a playstyle from a player's average skill and average daring. To set its thresholds, `app/tools/calibrate_styles.dart` plays seven simulated players through the app's own code (not the Python of the other experiments), with answers drawn by frequency from the 3,000 most common words. The players are: *best* (the guess with the smallest E every turn); *near-best* (one of the three best at random); *best-fit* (the best split among the words that still fit, so always an attempt at the word); *frequent* (the most common word that still fits); *random-fit* (any word that still fits); *wild* (a common word, ignoring the clues); and *wild-bold* (a rare word, ignoring the clues). Means are over the guesses made with more than one word left, the opening guess included. Table 5 gives English at five letters (200 games each); the output of this and three more settings is in `results/daring_calibration.txt`. That output was saved on 2026-10-24, before skill and luck were generalised to the answer law: skill and luck then assumed equal chances while daring's p used the frequency weights, and answers were drawn by frequency. The tables of this section have not been regenerated with the law-consistent scores.

Three things stand out. First, the best player has daring exactly 0, as Proposition 8 says. Second, daring is not independent of skill, and at the opening guess it is a function of skill alone (Proposition 8(5)): a player's daring there is fixed by the skill of the opener. After the opener it records what the player tried. Frequent and random-fit play a word that still fits on every turn whatever it costs in the split, and score about 20. Best-fit also always plays a word that fits, but the best-splitting one, so the cost factor is small and it scores 4. Near-best picks among the three best splits, which are often not possible words, and scores under 1. Wild and wild-bold ignore the clues: after the opener their guesses are rarely still possible and then have daring 0, so their 6.0 and 6.5 come mostly from the opener (this split was not measured separately). Third, daring is small for everyone, because the tries-to-spare and chance factors shrink it, so a good player seldom exceeds 10 (Table 6).

The levels depend on the setting. Table 7 gives the means in four settings. The picture is the same in all four: best-fit is at 3.5 to 4.8 (except at eight letters, where it plays almost as well as best, 0.5), and frequent and random-fit are at 14 to 24. Skill depends on length: the player who ignores the clues scores 48 at five letters and 69 at eight, since a longer word has more feedback patterns and a random guess splits better. A skill threshold set at five letters misplaces eight-letter play. The app's thresholds are a first calibration, set at five letters, and not measured on people.

skill of the guess	guesses	daring p10	p50	p90
0–40	830	0	0	19
40–55	620	0	0	42
55–70	901	0	28	33
70–80	468	0	18	23
80–90	361	0	10	14
90–100	377	0	1	6
100	1470	0	0	0

Table 6: Daring by the skill of the guess, all seven players pooled (English, five letters). The median falls as skill rises, since the cost factor does; the lower decile is zero because probes have no daring.

	en, 5	fr, 5	de, 4	en, 8
near-best	99.0 / 0.3	98.9 / 0.4	97.8 / 0.8	99.4 / 0.3
best-fit	91.4 / 4.0	87.1 / 4.8	88.7 / 3.5	99.0 / 0.5
frequent	66.3 / 19.7	63.7 / 20.7	62.6 / 20.2	77.1 / 16.7
random-fit	70.7 / 21.4	69.0 / 22.1	64.6 / 24.0	81.8 / 14.3
wild	48.2 / 6.0	45.3 / 5.5	41.5 / 7.3	68.9 / 7.0

Table 7: Mean skill / mean daring in four settings (200 games each, 100 in English at eight letters). The best player is 100/0 everywhere.

The app turns the two means into a playstyle: skill below 58, from 58 to 85, and 85 or more gives three tiers, and each tier splits by daring at 12, 12 and 2.5 respectively (loose, solid, sharp), so that *bold* means going for the word more than a player of that skill who works the clues. This gives six styles. The five-letter bots land as follows: best and near-best are in the sharp, careful style (analyst); best-fit in the sharp, bold one (maverick); frequent and random-fit in the solid, bold one (gambler); wild and wild-bold in the loose, careful one (wanderer). A margin of 3 points of skill and 1.5 of daring around the player's current style keeps one game either side of a line from changing it.

9.6 Do skill, luck and daring predict the tries?

If skill, luck and daring describe a game, they should predict how many tries it takes. Two tools test this. The first, `app/tools/predict_tries.dart`, plays 400 games per language by simulated players whose habits are drawn afresh for each game (the chance, per guess, of each of best, best-fit, frequent, random-fit and wild play), and fits the number of tries (a failed game counts as 7) by least squares on the game's mean skill, mean luck and mean daring over the guesses with a choice. Table 9 gives R^2. It should not be read as evidence, for three reasons. Answers were drawn by frequency while luck assumed equal chances, so the luck column is biased and not uncorrelated with skill (Section 4.1). The model is fitted and read on the same games, from one seed: the same model gave $R^2 = 0.542$ at 250 games and 0.483 at 400, so differences of a few points are within noise. And a lost game coded 7 is a censored value treated as exact, game means over guesses with a choice are endogenous (a short game averages few

guesses), and the last-try luck is the win indicator less its chance, so it nearly encodes the outcome of a game that reaches its last try.

The second, `app/tools/predict_tries3.dart`, replaces it. It scores every guess with the app's own code under an explicit answer law: **u** (equal chances, frequency off), **f** (by frequency) and **d** (the daily: equal chances over the 3,000 most common words). The same mixed-habit players play games whose answers are drawn by that law, and the tries are regressed on the means of skill, luck and daring. The fit uses the first half of the games and R^2 is read on the second half, which the fit never saw. The tool also prints how evenly each score fills its range (the share of all guesses in each tenth), the correlations between the three game means and their variance inflation factors, which show whether the coefficients can be read one at a time. It uses one seed. As in the first experiment, the last try's luck (outcome less win chance) enters the game means, so part of the luck R^2 is mechanical.

Results of the second experiment. English, five letters, 2,500 games per law (fit on 1,250, read on 1,250), one seed (`results/predict_tries3_{u,f,d}_en.txt`).

	u	f	d
mean tries (share lost)	4.67 (12.9%)	3.94 (5.4%)	4.37 (8.6%)
held-out R^2, skill alone	0.26	0.29	0.22
held-out R^2, luck alone	0.31	0.01	0.29
held-out R^2, daring alone	−0.01	0.01	0.00
held-out R^2, skill + luck	0.55	0.30	0.53
held-out R^2, skill + luck + daring	0.60	0.36	0.59
mean skill / luck / daring	74.8 / −0.8 / 11.3	81.5 / −0.5 / 8.1	76.3 / −0.2 / 11.1
VIF skill / luck / daring	1.40 / 1.04 / 1.42	1.49 / 1.00 / 1.49	1.46 / 1.04 / 1.50

Table 8: Held-out prediction of the tries a game takes, by answer law.

Three things follow. First, the mean luck is within one point of zero under every law, as Theorem 3 says for the law that draws the answer. Second, the three scores are close to uncorrelated as game means (the largest $|r|$ is the skill–daring pair, -0.52 to -0.57, which is the opening-guess identity of Proposition 8(5), not a defect); the variance inflation factors are all at most 1.5, so the coefficients can be read one at a time. Third, the scores jointly explain 36% of the variance of the tries when the answers are drawn by frequency and about 59% otherwise, and daring adds 4 to 6 points on top of skill and luck. Under law f luck explains almost nothing alone ($R^2 = 0.01$): a common word is both the likely answer and, on a bad split, the likely big pool, so the order of the pools by mass says little about how many guesses remain. The fitted line, the mean of the three, is the one the history chart shows ($9.38 - 0.0563\,\text{skill} - 0.0164\,\text{luck} - 0.0445\,\text{daring}$).

The spread is *not* even. Skill puts 42 to 56% of all guesses in its top tenth (most guesses are good); daring puts 58 to 71% in its lowest tenth (most guesses are not attempts at the word); luck is spread over its range but humped near the middle. These are properties of the players simulated, not of the definitions, but they mean that a score of 50 does not sit at the middle of the guesses played.

In Table 9, skill and luck together explain about half of the variance, and daring adds 1.5 to 4 points on top ($0.458 \to 0.483$, $0.538 \to 0.578$, $0.604 \to 0.619$), with a negative coefficient

regressors	en, 5	fr, 5	de, 4
skill	0.156	0.050	0.062
luck	0.286	0.425	0.490
daring	0.000	0.026	0.001
skill + luck	0.458	0.538	0.604
skill + daring	0.215	0.153	0.112
skill + luck + daring	0.483	0.578	0.619

Table 9: In-sample R^2 of the tries a game takes on its mean skill, luck and daring (400 simulated games each, answers by frequency, luck with equal chances, one seed; `results/predict_tries.txt`). Superseded by `predict_tries3`.

in every fit: given skill and luck, a player who went for the word more took fewer tries. Daring alone explains almost nothing, since by itself it does not say whether the attempt paid off. Whether the addition is real is what the held-out fit is for. The rest of the variance is the discreteness of the tries and what the game means lose (the number of guesses with a choice, and when the luck came). Luck explains more than skill alone here; a plausible reading, untested, is that a game that ends early had a lucky split, and part of it is mechanical (the last-try luck above).

The history page uses the fits to draw the expected guesses of a period. It takes the mean of the three fits of the second experiment, $9.38 - 0.0563\,\text{skill} - 0.0164\,\text{luck} - 0.0445\,\text{daring}$, and shows each term as what it does to a base. Skill, which is never negative, can only take guesses off; luck can do either; daring is measured from a typical level of 10 so that it too can add or take off. The weights are the coefficients of the fits above, from the first half of the games (only R^2 is read on the second half); one seed, English five letters, so a first reading, not measured on people.

9.7 A worked game

`worked_game.py` follows one game: CARET, then LEAST, against AGENT. After CARET 18 words remain. LEAST splits them into classes of $5, 3, 2, 2, 1, 1, 1, 1, 1, 1$. Its expected pool is $E = 2.67$. The best real guess is SEPAL, with $E^* = 1.56$ and classes $3, 2, 2$ and eleven singletons. LEAST scores 78; SEPAL scores 100.

The answer falls in the class of 5 under LEAST: reduction 1.85 bits = 2.75 (skill) $+0.28$ (risk) -1.19 (luck), percentile luck -72%. The best guess SEPAL has skill 3.53 and risk 0.15 bits. LEAST is a possible answer, so it finishes the game at once with chance $1/18 = 5.6\%$; SEPAL is not a possible answer and cannot. This is a gamble taken to finish in fewer guesses. It is priced in the risk term (0.28 against 0.15 bits), but only as a singleton class. The accounting gives a win no more credit than any other singleton, although a win saves a whole guess (Section 10).

9.8 The game's answer draw

Luck computed with equal chances has mean zero when each possible word is equally likely. The game draws answers by word frequency. `prior_bias.py` gives the mean first-guess luck, computed with equal chances, when the answer is drawn by frequency: $+10.2\%$ for TARES,

+10.6% for CRANE, +7.6% for SLATE, −6.4% for ABBEY, −8.2% for MAMMA. Restricting the draw to the 500 most common words changes these by about one point. Common words tend to fall in small classes of strong guesses and large classes of weak ones. Under (U) the means are zero to machine precision (Table 3). This cross-law bias is what Section 4.1 describes; the app now computes luck with the law the answer is drawn by, for which the mean is exactly zero (Theorem 3(3)). The opposite direction (luck by frequency when answers are equal) was not measured.

9.9 A bound on the best conceivable split

Three bounds on the number of groups were tested: the naive $\min(N, 3^\ell)$; a bound B counting 2 outcomes at positions fixed by the clues and 3 elsewhere; and a bound C taking, over all real guesses, the product over letters of how many of green/yellow/grey that letter could show against the words still possible. On 14 simulated games (40 positions after 0, 1 or 2 guesses), B and C equal the naive bound in every position. Where $N > 243$ the best scanned real guess made about 63% of the allowed groups. Where $N \le 243$ a real guess separated every word in 3 of 20 positions. C is not a valid bound for repeated letters (7 of 109,782 guesses exceeded it, all with a repeated letter). The bounds do not give a reachable best, so skill is normalised by the best real guess.

9.10 Numerical check of the theorems

`verify_theorems.py` checks Theorems 3 and 4 and Proposition 5 independently of the app, under (U). On 20,000 random partitions, luck lies strictly in $(-1, 1)$, is strictly monotone in the pool, is unchanged by scaling the classes, has $|\text{mean}| < 2 \times 10^{-16}$, and has variance equal to $(1 - \sum q^3)/3$ to the same precision. On 1,500 simulated games of four guesses, with uniform answers and guesses depending only on the past, the mean luck of each guess is within sampling error of zero (for the first guess +0.008, standard deviation 0.57). All correlations of a guess's luck with the skill of any guess so far, and with earlier luck, are within four standard errors of zero (largest in magnitude 0.084, tolerance 0.103); the "skill" there is the linear $(N-E)/(N-1)$, a number fixed by the past, which is all the theorem needs.

`verify_laws.py` checks the weighted-law statements. With rational arithmetic, on 3,000 random weighted partitions (ties included) luck has mean exactly 0, variance $(1 - \sum q^3)/3$, lies strictly in $(-1, 1)$ and is strictly monotone; luck under a different law has a nonzero mean; the tie rule with weight λ has mean $(2\lambda - 1)\sum q^2$; the rescaled luck has mean 0 for $\{2, 2\}$ and $-\frac{1}{2}$ for $\{3, 1\}$; the last-try luck has mean 0 and the daring bound of Proposition 8(4) holds. On 800 simulated games of three guesses by a randomising player, with answers drawn by frequency from the 250 most common words and luck computed with that law, the mean luck of each guess is within four standard errors of zero and all correlations with the skill so far and with earlier luck are within tolerance (`results/verify_laws.txt`; the skill there uses a sample of 50 guesses for E^*, again a number fixed by the past).

10 Open problems

1. **Non-uniform priors.** *Answered* (Section 2.1): replace sizes by effective sizes $s_i = Nq_i$, so $E = N\sum_i q_i^2$; Theorems 3 and 4 hold unchanged, pools are ranked by mass, and luck

is exactly unbiased under the law used (and biased under another, Section 4.1). What remains open is whether a player can be assumed to know the frequencies.

2. **Lookahead.** Skill scores the split made now. A guess can score lower and play better over the rest of the game. Is there a skill that credits lookahead and depends only on the player's knowledge, for example through the expected number of further guesses of an optimal continuation [4]?

3. **Scale of skill.** The logarithm of the expected pool is used because the linear scale compresses first guesses near 100. Other Rényi orders, or a scale calibrated to the guesses a strong player needs, are alternatives. Test against how strong players rank openings.

4. **Risk attitude and tries left.** The best entropy order should depend on the guesses remaining: an even split with many left, a gamble with few (Section 9.4). Is there a skill that depends on the tries left, for example the chance of finishing within them under an optimal continuation?

5. **Credit for finishing now.** A guess that is a possible answer wins at once with chance $p(g)$ ($1/N$ under (U)); on the last try this is now the skill (Proposition 6), but not before it. In the accounting the winning class is a singleton like any other, with reduction $\ln N$. But landing in another singleton still costs one more guess; landing in the winning class does not. Skill and risk therefore undercount guesses that can win. A measure based on the expected number of remaining guesses would count it. How should this be combined with the entropy terms?

6. **Total luck of a game.** Is there a normalised total luck comparable across games of different length, for example $\sum_t \Lambda_t / \sqrt{\sum_t \operatorname{Var} \Lambda_t}$?

7. **Relay-race credit.** How to credit the head start of a board to the earlier guesses that produced it.

8. **Ties.** *Answered* (Section 4.1): a rule giving tied pools weight λ has mean $(2\lambda - 1) \sum_i q_i^2$, so within that family the mid-rank $\lambda = \frac{1}{2}$ is the only one that is fair for every law. Rules outside the family are not covered.

9. **Hard modes, other feedback.** Only the following is used about f: a map to a finite set with all-green exactly at equality. *Answered, largely trivially*: in a game where later guesses must obey earlier clues, every word still possible obeys them (the answer does), so $\Omega \subseteq \mathcal{G}$ still holds and all results go through with $\mathcal{G}$ replaced by the legal guesses; only E^* can rise, because the probes are no longer available. How much it rises is not measured.

10. **Playstyles on real players.** The thresholds of Section 9.5 come from simulated players at five letters. They should be set on real play, and probably per length: skill is not comparable across lengths (the player who ignores the clues scores 48 at five letters and 69 at eight). Is there a normalisation of skill and daring, for example by their distribution over a reference set of players, that makes one threshold fit all?

11. **The form of daring.** D is a product of four factors chosen to be monotone in what a bold choice means. Other forms (a sum, other powers, the chance in bits rather than

as a probability) would fit the tries better or worse. Which form predicts the tries best, and does a fitted form beat the product?

12. **Approximating E^*.** The exact scan is quadratic in the list size. Are there provable shortcuts, such as a lower bound on E^* from letter statistics of Ω, for very large lists or scripts with more patterns?

11 Reproducibility

The app computes these quantities in

- `app/lib/logic.dart`: `scoreGuess`, `chanceLuck`, `skillAgainst`, `analyseGuesses`, `splitRisk` and `daringOf` (risk and daring);
- `app/lib/multi_analysis.dart`: `analyseMulti`;
- `app/lib/playstyle.dart`: the tiers, the splits and the margins of Section 9.5.

Tests are in `logic_test.dart`, `review_plots_test.dart`, `multi_analysis_test.dart` and `playstyle_test.dart` under `app/test`: bounds, zero mean over 500 random partitions, the plot geometry against the table, the properties of risk and daring, and where the simulated players of Table 7 fall. The simulated players of Section 9.5 are run with `dart run tools/calibrate_styles.dart [code] [length] [games]` from `app`, which writes to the console; the saved output is `results/daring_calibration.txt`. The tries regression of Section 9.6 is `dart run tools/predict_tries.dart [code] [length] [games]`, saved as `results/predict_tries.txt`; its replacement is `dart run tools/predict_tries3.dart [law] [code] [length] [games]` with the law `u`, `f` or `d`, saved as `results/predict_tries3_`⟨law⟩`_en.txt`. The other experiments are in `docs/paper/experiments`, share `common.py`, and each states its command and output file:

- `first_guess.py`, `first_guess_summary.py`: first guesses;
- `luck_definitions.py`, `prior_bias.py`: luck definitions, effect of the answer draw;
- `play_skill.py`, `risk_attitude.py`: simulated players, the risk question;
- `reduction_accounting.py`, `worked_game.py`: skill, risk and luck terms, and one worked game;
- `capacity_bound.py`: the negative result;
- `verify_theorems.py`, `verify_laws.py`: numerical checks (the second for weighted laws, ties, the last try and the daring bound).

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